Causative agents of human malaria:
The organisms: Four species of Plasmodium cause malaria in humans.

- *Plasmodium vivax:* (Benign Tertian malaria)
- *Plasmodium falciparum:* (Malignant tertian malaria)
- *Plasmodium malaria:* (Benign Quartan malaria)
- *Plasmodium ovale:* (Benign tertian malaria)

The two most common species are P. vivax and P. falciparum, WHO reports (2018) that falciparum being the most pathogenic of all.

8.5.1 Geographical Distribution

Malarial parasites are found in all countries. In India, malaria continues to be a major public health threat.

8.5.2 Habitat

The malarial parasites infecting man, after passing through a developmental phase in the parenchyma cells of the liver, reside inside the red blood corpuscles and are carried by the circulating blood to all the organs.

8.5.3 Vectors

Human malaria is transmitted by over 60 species of female Anopheles mosquito.

Human malarial parasite – *Plasmodium falciparum*

Of all the human malaria parasites, *P. falciparum* is the most highly pathogenic and responsible for malignant tertian malaria. This is a form of disease which runs an acute course in non-immune patients and is frequently fatal if untreated.

8.5.4 Life Cycle



The malaria parasite passes its life cycle in two different hosts and comprises of two phase as follows,

Definitive host: Female Anopheles mosquito (a sexual phase of parasite occurs).

Intermediatehost: Human (an asexual phase of parasite occurs).

Thus, life cycle of malaria parasite show alternation of generations- asexual and sexual generation in two different hosts (Figure 8.12).

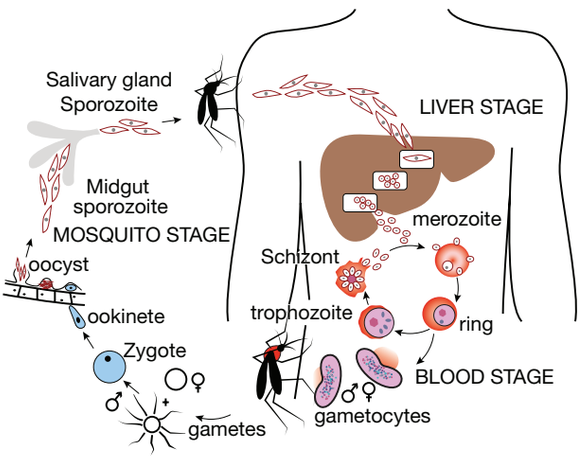


Figure 8.12: Lifecycle of Plasmodium spp

8.5.5 Human Cycle (Asexual Phase – Schizogony)

Human infection occurs when the sporozoites (the infective forms of the parasite are present in the salivary gland of the mosquito) are injected into blood capillaries when the mosquito feeds on blood after piercing the skin. The malarial parasite multiplies by division and the process designated as Schizogony (schizo: to split, gone: generation).

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